Manishika Balamurugan

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Education

University of Michigan - Ann Arbor — Expected May 2027
Bachelor of Science in Computer Science & Biopsychology, Cognition, and Neuroscience. ***(GPA: 3.91/4.00)*** — *Ann Arbor, MI*

- **Honors:** Rogel Scholar (Full Ride), William J. Branstrom Prize (Top 5%), University Honors

Experience

Sartorius — Jan 2026 – May 2026, Aug 2026 – Dec 2026
Student Software Engineer, UM CoE Multidisciplinary Design Program — *Ann Arbor, MI*

- Designing containerized backend systems (Docker, Node.js) for remote lab monitoring and control.
- Building secure REST APIs and a React dashboard for real-time telemetry and configuration
- Collaborating with industry engineers to align architecture with commercial products.
- Integrating a hardware simulator to support development and automated testing without physical devices.

American Airlines Sponsored Project — Jan 2026 – Present
Student Machine Learning Engineer, Michigan Data Science Team — *Ann Arbor, MI*

- Developing a predictive system to minimize high-cost flight simulator cancellations using historical logs.
- Building a hybrid ML pipeline integrating Random Forests with a GenAI layer to extract features from unstructured pilot notes.
- Partnering with stakeholders to deploy production-ready models for optimized training scheduling.

Michigan Daily — Sep 2025 – present
Web Developer — *Ann Arbor, MI*

- Redesigned frontend gameplay components for Ann Arbor Geuguessr game using Svelte and TypeScript.
- Built a SQL-backed daily game generation system used by thousands of readers.

Projects

Replica of GPT-2 | ***PyTorch***

- Implemented a GPT-2–style Transformer from scratch, including multi-head self-attention and positional encodings.
- Built a custom training loop, optimizer, and evaluation pipeline.

Brain Tumor Detection & Classification | ***PyTorch***

- Achieved 99.7% binary and 95.7% multi-class accuracy with CNNs (ResNet34, ConvNeXt).
- Applied transfer learning, augmentation, and evaluation metrics.

AI-Powered Student Life Dashboard | ***FastAPI, JavaScript, React***

- Built a full-stack web application using FastAPI and React with NLP-based AI assistants.
- Designed REST APIs and frontend state management for real-time user interactions.

Research

Computational & Cognitive Neuroscience Lab, University of Michigan — Aug 2024 – Present
Research Assistant — *Ann Arbor, MI*

- Built and maintained R-based data processing pipelines for MRI datasets, ensuring reproducibility and accuracy.
- Conduct preprocessing, quality control, and statistical analysis for hundreds of samples.
- Trained an internal AI assistant to streamline literature review and lab communication workflows.

Regenerative Imaging Laboratory, University of Pittsburgh — May 2025 – Aug 2025
Research Fellow — *Pittsburgh, PA*

- Analyzed hippocampal connectivity in 18 brains using diffusion MRI and tractography software.
- Wrote MATLAB scripts for clustering neural connections and modeling network patterns.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, C, R
Frameworks/Libraries: React, Flask, Node.js, FastAPI, PyTorch, Svelte, scikit-learn, OpenCV
Tools: Docker, Git, Jupyter, REST APIs, MATLAB
ML/DS: CNNs, Transformers, Transfer Learning, Data Pipelines